SOFTWARE REQUIREMENT SPECIFICATION (SRS)

NE Brothers & MAKU Security Workforce Management System

Version: 3.0 (Advanced – Geo-Fencing + Face Recognition)
Scope: Phase 1 MVP with Biometric & Location Intelligence

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1. Introduction

1.1 Purpose

This document provides a comprehensive and detailed definition of the system requirements for the **Security Workforce Management System**, designed specifically for organizations such as NE Brothers and MAKU Security.

The purpose of this system is to:

- Digitize workforce operations
- Ensure **accurate and tamper-proof attendance**
- Provide **real-time operational visibility**
- Enhance **employee accountability and safety**
- Deliver **transparent reporting to clients**

This document serves as:

- A **blueprint for system design and development**
- A **reference for stakeholders and decision-makers**
- A **baseline for quality assurance and testing**

1.2 Scope

The system addresses critical operational challenges in traditional security workforce management, including:

- Proxy or fraudulent attendance
- Lack of real-time monitoring
- Delayed incident reporting

- Limited client visibility

To resolve these issues, the system integrates:

1. Geo-Fencing (Location Verification)

Ensures employees are physically present within a defined site boundary.

2. Face Recognition (Identity Verification)

Ensures the person marking attendance is the **actual assigned employee**, preventing impersonation.

3. Centralized Monitoring

Provides admins and clients with real-time operational insights.

1.3 Stakeholders

1.3.1 Admin (Company Management)

Admins are responsible for system configuration and oversight. Their responsibilities include:

- Managing employees, clients, and sites
- Configuring geo-fencing rules and radius
- Monitoring attendance and violations
- Reviewing incidents and generating reports

1.3.2 Employees (Security Guards)

Employees are operational users who:

- Mark attendance using mobile/web
- Submit incident reports
- Respond to assigned duties

1.3.3 Clients

Clients are external stakeholders who:

- Monitor guards assigned to their sites
- Review attendance and compliance
- Access reports and incident logs

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2. System Overview

The system is composed of four integrated modules:

2.1 Employee Mobile Application

The primary interface for field operations. It enables:

- **Geo + Face verified attendance**
- Incident reporting with media
- SOS emergency alerts

This module ensures **real-time, on-ground data capture**.

2.2 Employee Web Portal (Fallback System)

Provides a browser-based alternative when:

- Mobile device is unavailable
- App is not functioning

Ensures **operational continuity without disruption**.

2.3 Admin Web Panel

The central control system that allows:

- Configuration of geo-fencing and face recognition
- Monitoring of workforce activity
- Management of system data

2.4 Client Web Panel

A restricted interface providing:

- Visibility into assigned guards
- Attendance and compliance status
- Incident and report access

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3. Core Module: Smart Attendance System

This module combines **Geo-Fencing + Face Recognition** to ensure **dual-layer verification**.

3.1 Geo-Fencing (Location Validation)

3.1.1 Concept

Geo-fencing defines a **virtual boundary** around a physical site. Attendance is only valid if the employee is inside this boundary.

3.1.2 Site Configuration

Each site includes:

- Latitude and Longitude (fixed location)
- Radius (admin-defined boundary)
- Client association

3.1.3 Radius Configuration

Admins can define radius based on site type:

- Small sites → 30–50 meters
- Medium sites → 50–100 meters
- Large sites → 100–200 meters

This ensures flexibility and accuracy.

3.1.4 Validation Process

1. Capture employee GPS location
2. Calculate distance from site
3. Compare with radius

Outcome:

- Inside → Valid
- Outside → Block or Flag

3.1.5 Policy Modes

Strict Mode

- Attendance blocked outside boundary

Flexible Mode

- Attendance allowed but flagged for review

3.2 Face Recognition (Identity Validation)

3.2.1 Concept

Face recognition ensures that the person marking attendance is the **authorized employee**, eliminating impersonation risks.

3.2.2 Enrollment Process

Before using the system:

- Employee face is registered
- Multiple angles/images may be captured

- Stored securely in system

3.2.3 Attendance Verification Process

During check-in:

1. Camera captures live image
2. System compares with stored face data
3. Generates similarity score

3.2.4 Decision Logic

- Match above threshold (e.g., 90%) → Valid
- Match below threshold → Rejected or flagged

3.2.5 Anti-Fraud Measures

- Detect static images (photo spoofing)
- Require live capture (blink/movement detection – future enhancement)

3.2.6 Benefits

- Prevents proxy attendance
- Ensures identity authenticity
- Strengthens system integrity

3.3 Combined Validation Workflow

Attendance is only considered **fully valid** when:

- Location is within geo-fence AND
- Face recognition is successful

Validation Outcomes

Condition	Result
Geo ✓ + Face ✓	Valid Attendance
Geo ✗ + Face ✓	Out-of-Zone
Geo ✓ + Face ✗	Identity Failure
Geo ✗ + Face ✗	Rejected

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4. Functional Requirements

4.1 Employee Mobile Application

4.1.1 Authentication

Description:

Provides secure access to the system.

Features:

- Credential-based login
- Company selection
- Session persistence

Outcome:

- Ensures only authorized users access system

4.1.2 Duty Check-In (Geo + Face Verified)

Detailed Workflow

1. User taps “Start Duty”

2. System activates GPS
3. Retrieves assigned site details
4. Performs geo-fence validation
5. Opens camera
6. Captures live image
7. Performs face recognition
8. Stores:
    - Location
    - Image
    - Match score
    - Timestamp

Validation Rules

- GPS must be enabled
- User must be within allowed radius
- Face must match registered identity

Outputs

- Valid
- Out-of-zone
- Face mismatch
- Rejected

4.1.3 Duty Check-Out

Description:

Marks end of shift.

Process:

- Capture GPS
- Record timestamp

Enhancements:

- Detect early departure
- Compare with expected shift

4.1.4 Shift Management

Employees can:

- View assigned shifts
- Access site details
- Use navigation assistance

4.1.5 Incident Reporting

Description:

Allows reporting of events or anomalies.

Input:

- Category
- Description
- Image

System Adds:

- Geo-location
- Timestamp
- Site reference

Outcome:

- Immediate visibility to admin

4.1.6 SOS Emergency Feature

Description:

Provides immediate alert in emergency situations.

Process:

- Press and hold SOS button
- Send high-priority alert

Data Sent:

- Live location
- Employee details

Outcome:

- Enables rapid response

4.2 Employee Web Portal

4.2.1 Authentication

Same as mobile login.

4.2.2 Web Check-In

- Geo-fencing validation
- Camera-based face verification

4.2.3 Limitations

- Browser dependency
- Lower reliability than mobile

4.3 Admin Web Panel

4.3.1 Dashboard

Displays:

- Workforce summary
- Attendance status
- Alerts and incidents

4.3.2 Geo-Fence Management

Admins can:

- Define site location
- Configure radius
- Assign employees

4.3.3 Face Recognition Management

Admins can:

- Register employee faces
- Update biometric data
- Review failed attempts

4.3.4 Attendance Monitoring

Admins can view:

- Location accuracy
- Face match score
- Attendance flags

4.3.5 Reporting

Reports include:

- Attendance logs
- Geo violations
- Face mismatches

4.4 Client Web Panel

4.4.1 Dashboard

Shows:

- Guard status
- Attendance indicators

4.4.2 Geo + Identity Status

Clients can see:

- Whether guard is within site
- Whether identity is verified

4.4.3 Reports

Clients can download:

- Attendance reports
- Incident summaries

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5. Non-Functional Requirements

5.1 Performance

- Fast validation (few seconds)
- Real-time updates

5.2 Security

- Encrypted data storage
- Secure biometric handling

5.3 Reliability

- High uptime
- Error handling

5.4 Usability

- Simple interface
- Minimal training

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6. Constraints

- Requires GPS, camera, and internet
- Dependent on device capability

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7. Future Enhancements

- Liveness detection (anti-spoofing)
- Offline attendance sync

- AI-based analytics

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8. Conclusion

This system provides a **highly reliable, secure, and intelligent workforce management solution** by combining:

- **Geo-Fencing (location validation)**
- **Face Recognition (identity validation)**

This dual-layer verification ensures:

- Accurate attendance
- Fraud prevention
- Operational transparency
- Client confidence